Purpose

This guide covers all replacement-meter charge scenarios supported by WaterWise. Each scenario has its own table and worked example. The server applies the final calculation and the record-entry screen shows the carry basis used.

Core formula

Recorded consumption = carried old-meter usage + new-meter usage

New-meter usage = new meter reading - new meter baseline. The minimum charge is a billing amount and is separate from recorded water consumption.

Decision rules

Old closing reading is higher	Use old closing reading - latest recorded reading. This takes priority over date rules.
Same day	Use the previous completed consumption.
1-7 days after latest reading	Carry 0 m3.
8-15 days after latest reading	Use the configured minimum charge volume.
16 days or more	Use the average of three valid historical months, rounded up to a whole number.
No water was used	Set the new meter reading equal to its baseline; record 0 m3 and bill the minimum charge.

Base values used in the examples

Latest recorded old-meter reading: September 23, 2026 = 99 m3. Historical valid consumption: July = 12 m3, August = 17 m3, September = 20 m3. Three-month average: 16.33 m3, rounded up to 17 m3. Configured minimum volume: 5 m3.

Scenario 1 ? Positive old-meter difference

Item	Example / Result
Latest recorded reading	**80 m3 on September 23**
Old meter closing reading	99 m3
Incident effective date	September 27, 2026
Date gap	4 days
Priority rule	Old closing reading is higher; difference takes priority.
Old-meter carry	99 - 80 = 19 m3
New baseline / reading	0 m3 / 0 m3
Recorded consumption	19 + (0 - 0) = 19 m3
Billing result	**Tiered bill based on 19 m3**

Scenario 2 ? Same-day replacement

Item	Example / Result
Latest reading	**20 m3 consumption on September 23**
Incident effective date	September 23, 2026
Date gap	0 days
Rule	Use previous completed consumption.
Old-meter carry	20 m3
New baseline / reading	0 m3 / 0 m3
Recorded consumption	20 + (0 - 0) = 20 m3
Billing result	**Tiered bill based on 20 m3**

Scenario 3 ? 1 to 7 days: no carried usage

Item	Example / Result
Latest reading date	**September 23, 2026**
Incident effective date	September 27, 2026
Date gap	4 days
Rule	Carry 0 m3.
Old-meter carry	0 m3
New baseline / reading	0 m3 / 0 m3
Recorded consumption	0 + (0 - 0) = 0 m3
Billing result	**Minimum charge**

Scenario 4 ? 8 to 15 days: minimum-charge window

Item	Example / Result
Latest reading date	**September 23, 2026**
Incident effective date	October 2, 2026
Date gap	9 days
Rule	Use configured minimum charge volume.
Configured minimum volume	5 m3
New baseline / reading	0 m3 / 0 m3
Recorded consumption	5 + (0 - 0) = 5 m3
Billing result	**Minimum charge**

Scenario 5 ? 16 days or more: three-month average

Item	Example / Result
Latest reading date	**September 23, 2026**
Incident effective date	October 12, 2026
Date gap	19 days
Valid historical months	12 m3, 17 m3, and 20 m3
Average calculation	(12 + 17 + 20) / 3 = 16.33 m3
Rounded-up carry	CEIL(16.33) = 17 m3
New baseline / reading	0 m3 / 0 m3
Recorded consumption	17 + (0 - 0) = 17 m3
Billing result	**Tiered bill based on 17 m3**

Scenario 6 ? New meter has actual usage

Item	Example / Result
Incident carry basis	**Minimum-charge window; carried usage = 5 m3**
New meter baseline	0 m3
New meter reading	3 m3
New-meter usage	3 - 0 = 3 m3
Recorded consumption	5 + 3 = 8 m3
Billing result	**Tiered bill based on 8 m3**

Scenario 7 ? New meter starts above zero

Item	Example / Result
Incident carry basis	**Minimum-charge window; carried usage = 5 m3**
New meter baseline	20 m3
New meter reading	23 m3
New-meter usage	23 - 20 = 3 m3
Recorded consumption	5 + 3 = 8 m3
Billing result	**Tiered bill based on 8 m3**

Scenario 8 ? No water was used

Item	Example / Result
Reader action	**Select ?No water was used?**
New meter baseline	0 m3
Recorded new meter reading	Automatically set to 0 m3
New-meter usage	0 - 0 = 0 m3
Carried usage	0 m3
Recorded consumption	0 + 0 = 0 m3
Billing result	Minimum charge only
Safety behavior	**Reading field becomes read-only after selection**

Scenario 9 ? Invalid new meter reading

Item	Example / Result
New meter baseline	**20 m3**

Entered new meter reading	18 m3
Validation result	Rejected: reading cannot be lower than the baseline.
Required action	Verify the meter value or correct the replacement incident.
Consumption / billing	**No record or bill is created until corrected.**

Scenario 10 ? No valid three-month average

Item	Example / Result
Date gap	**16 days or more**
Historical data	Fewer than three valid actual months
Fallback rule	Use configured minimum charge volume.
Example carried usage	5 m3
New baseline / reading	0 m3 / 0 m3
Recorded consumption	5 m3
Billing result	**Minimum charge**

Meter reader sequence

1. Record the replacement incident with the effective date, old meter closing reading, and new meter baseline.
2. Open Record Consumption and enter the new meter reading.
3. If the resident confirms that no water was used, select "No water was used".
4. Review the displayed basis, recorded consumption, and estimated bill.
5. Confirm and record. The system stores the audit basis and creates the bill.

Important notes

The minimum charge is a billing amount, not water consumption. A no-usage record stores 0 m3 while still charging the configured minimum amount. Positive old-meter differences take priority over the date window. The updated database RPC must be deployed in Supabase for server-side behavior to be active.